Learning Management System

Group Members:

- Tuba
- Emaan Fatima
- Ayesha Hassan

Functional Requirements

User Registration
• The system must allow students and teachers to register.
• The system must send a registration confirmation notification.

2. User Login
• Students, teachers, and admin must be able to log in.
• The system must send login confirmation notifications.

3. Course Management (for Admin & Teachers)
• Admin must be able to create and manage courses.
• Admin must approve teacher accounts.
• Teachers must upload course material and assignments.

4. Enrollment
• Students must be able to enroll in courses.
• The system must record enrollment information.

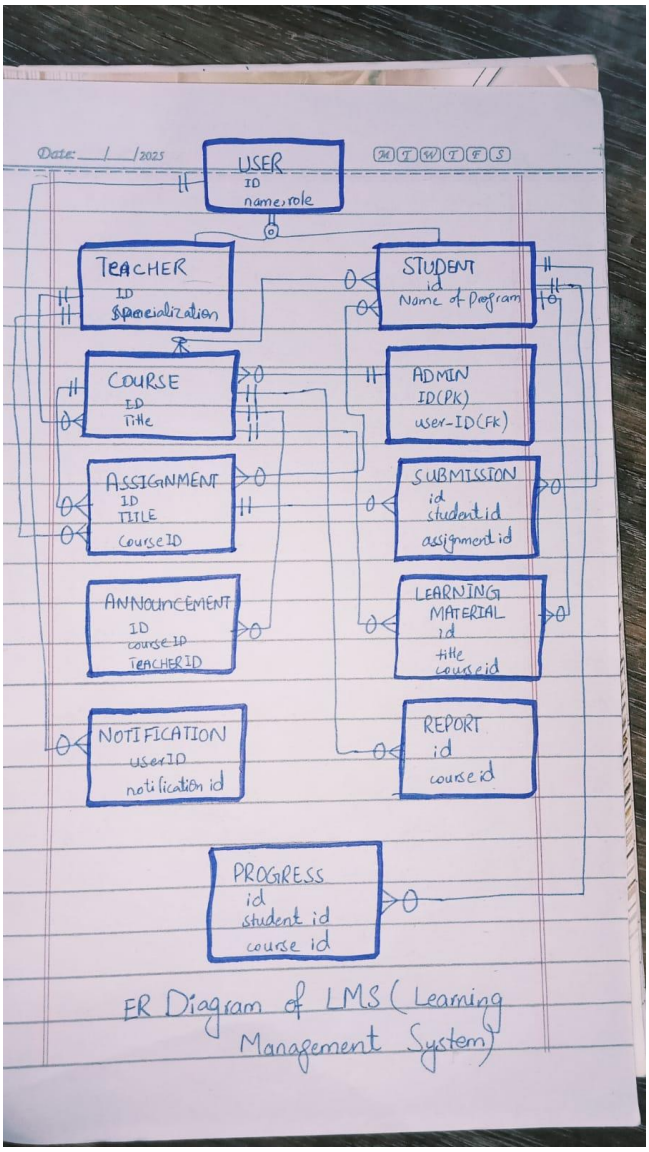
5. Assignments & Quizzes
• Students must be able to submit assignments.
• Students must be able to attempt quizzes.
• Teachers must submit grades.

6. Attendance Management
• Teachers must be able to update attendance.

7. Notifications
The system must send notifications for:
• registration

• login
• new material
• grades
• submissions

8. Reports & Logs (Admin)
• Admin must be able to view system logs.
• Admin must generate reports.



CREATE TABLE User (

```
    user_id INT PRIMARY KEY,
    name VARCHAR(50),
    role VARCHAR(20)
);

CREATE TABLE Teacher (
    teacher_id INT PRIMARY KEY,
    specialization VARCHAR(50),
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES User(user_id)
);

CREATE TABLE Student (
    student_id INT PRIMARY KEY,
    program VARCHAR(50),
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES User(user_id)
);

CREATE TABLE Admin (
    admin_id INT PRIMARY KEY,
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES User(user_id)
);

CREATE TABLE Course (
    course_id INT PRIMARY KEY,
```

```
    title VARCHAR(100),
    teacher_id INT,
    FOREIGN KEY (teacher_id) REFERENCES Teacher(teacher_id)
);

CREATE TABLE Assignment (
    assignment_id INT PRIMARY KEY,
    title VARCHAR(100),
    course_id INT,
    FOREIGN KEY (course_id) REFERENCES Course(course_id)
);

CREATE TABLE Submission (
    submission_id INT PRIMARY KEY,
    student_id INT,
    assignment_id INT,
    FOREIGN KEY (student_id) REFERENCES Student(student_id),
    FOREIGN KEY (assignment_id) REFERENCES Assignment(assignment_id)
);

CREATE TABLE LearningMaterial (
    material_id INT PRIMARY KEY,
    title VARCHAR(100),
    course_id INT,
    FOREIGN KEY (course_id) REFERENCES Course(course_id)
);
```

```
CREATE TABLE Announcement (
    announcement_id INT PRIMARY KEY,
    course_id INT,
    teacher_id INT,
    FOREIGN KEY (course_id) REFERENCES Course(course_id),
    FOREIGN KEY (teacher_id) REFERENCES Teacher(teacher_id)
);

CREATE TABLE Notification (
    notification_id INT PRIMARY KEY,
    user_id INT,
    FOREIGN KEY (user_id) REFERENCES User(user_id)
);

CREATE TABLE Report (
    report_id INT PRIMARY KEY,
    course_id INT,
    FOREIGN KEY (course_id) REFERENCES Course(course_id)
);

CREATE TABLE Progress (
    progress_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    FOREIGN KEY (student_id) REFERENCES Student(student_id),
    FOREIGN KEY (course_id) REFERENCES Course(course_id)
);
```

```
INSERT INTO User VALUES (1, 'Ali', 'Student');
INSERT INTO User VALUES (2, 'Sara', 'Teacher');
INSERT INTO Student VALUES (101, 'BSAI', 1);
INSERT INTO Teacher VALUES (201, 'AI', 2);
INSERT INTO Course VALUES (301, 'Database', 201);
INSERT INTO Assignment VALUES (401, 'SQL Task', 301);
INSERT INTO Submission VALUES (501, 101, 401);
INSERT INTO LearningMaterial VALUES (601, 'Lecture 1', 301);
INSERT INTO Announcement VALUES (701, 301, 201);
INSERT INTO Notification VALUES (801, 1);
INSERT INTO Report VALUES (901, 301);
INSERT INTO Progress VALUES (1001, 101, 301);
```